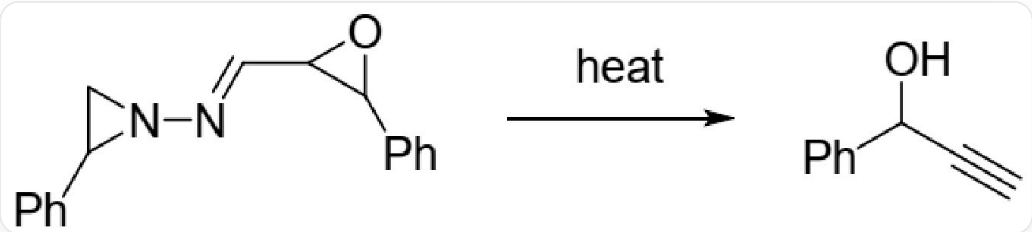
Question

In 1995, Professor Kim's research group reported the following reaction:



N1(CC1C2=CC=CC=C2)/N=C/C3C(C4=CC=CC=C4)O3 yields C#CC(C1=CC=CC=C1)O under heating conditions

Mechanistic studies have shown that, under heating conditions, the substrate molecule first decomposes to give $\mathbf{A}$ and $\mathbf{B}$, the molecular weight of $\mathbf{A}$ is approximately 104, $\mathbf{B}$ undergoes ring-opening to give $\mathbf{C}$, $\mathbf{C}$ undergoes intramolecular proton transfer to give $\mathbf{D}$, $\mathbf{D}$ then releases a molecule of gas to give $\mathbf{E}$, and $\mathbf{E}$ undergoes a 1,2-H shift to give the product.

The following statements are made:

1. The degree of unsaturation of $\mathbf{A}$ is 5

2. $\mathbf{B}$ is a 1,3-dipole

3. $\mathbf{C}$ has a charge-separated structure and the types of atoms bearing the positive and negative formal charges are different

4. The gas released by $\mathbf{D}$ is not a common gas in the atmosphere

5. All atoms in $\mathbf{E}$ (except hydrogen atoms) satisfy the octet rule

Which of the following options contains the most correct statements?

A. All statements are incorrect.

B. 1,2,3

C. 2,3,4

D. 3,4,5

E. 4,5

F. 2,3,4,5

G. 1,2

H. 1,3

I. 1,5

J. 2,4

K. 2,3,5

L. 2,5

M. 3,4

N. 2,4,5

O. 1,2,4

P. 1,3,5

Q. 2,3,4

Answer

Correct Answer: **B**

Detailed Explanation

The 2-phenylaziridine ring in the substrate easily undergoes a ring-opening reaction upon heating, eliminating styrene. The molecular weight of styrene is exactly 104, therefore $\mathbf{A}$ is styrene.

CHECKPOINT 1 PTS

Elimination of the 2-phenylaziridine ring produces $\mathbf{A}$, $\mathbf{A}$ is styrene, with an unsaturation degree of 5, statement 1 is correct

The structure of the remaining $\mathbf{B}$ is `[N-]=[N+]=CC1C(C2=CC=CC=C2)O1`, which is a 1,3-dipole.

CHECKPOINT 1 PTS

The structure of $\mathbf{B}$ is `[N-]=[N+]=CC1C(C2=CC=CC=C2)O1`, which is a 1,3-dipole, statement 2 is correct

$\mathbf{B}$ undergoes ring-opening, and the negative charge on the nitrogen attacks and opens the epoxide ring, yielding $\mathbf{C}$: `[O-]C(C1=CC=CC=C1)/C=C\[N+]#N`, with formal charges located on nitrogen and oxygen, respectively.

CHECKPOINT 1 PTS

The structure of $\mathbf{C}$ is `[O-]C(C1=CC=CC=C1)/C=C\[N+]#N`, with formal charges located on nitrogen and oxygen, respectively, statement 3 is correct

$\mathbf{C}$ undergoes intramolecular proton transfer to yield $\mathbf{D}$. The oxygen anion can abstract the hydrogen on the carbon atom attached to the diazo group, yielding `OC(C1=CC=CC=C1)/C=[C-]\[N+]#N`. Under heating conditions, the diazo group can decompose to produce nitrogen gas, which is the gas produced in the step from $\mathbf{D}$ to $\mathbf{E}$.

CHECKPOINT 1 PTS

$\mathbf{D}$ releases nitrogen gas when forming $\mathbf{E}$, which is a common gas in the atmosphere, statement 4 is incorrect

After the departure of nitrogen gas, carbene $\mathbf{E}$ is generated in situ, with the structure `OC(C1=CC=CC=C1)C=[C]`. Subsequently, a hydrogen migration occurs, with an alkenyl hydrogen migrating to the carbene.

CHECKPOINT 1 PTS

$\mathbf{E}$ contains a carbene, and the carbene carbon atom does not satisfy the octet rule, statement 5 is incorrect

Therefore, the correct statements are 1, 2, and 3. Choose option B.